Instructions for handling an RMA unit

Preliminary steps:
This process should be followed for all RMA units. At the incoming inspection stage, make sure that the units have been returned in a suitable package.
A visual inspection should be performed to ensure that the unit was not damaged during transportation.
At this stage the units should not be mixed with the current production units, separate shelves should be used.
Open an RMA identity card that includes the following:

1. RMA number
2. Part Number
3. Serial Number
4. Customer name
5. Customer complaint document

Open an RMA traveler card that includes the following:

1. RMA number
2. Part Number
3. Serial Number
4. Customer name

Performing incoming tests and rework

1. Quality visual check for the units before opening the cover. Document the finding by pictures and writing on the RMA identity card.
2. Failure validation by test engineer to verify the customer complaint.
3. Open the cover
4. Quality visual check or internal assembly. Document the finding by pictures and writing on the RMA identity card.
5. Test by test engineer, resistance, negative voltage, enable the amplifier and test for performance. Write the finding on the RMA identity card.
6. Engineering look at the results and the test technician assumptions for the failure reason and recommendation for fixing the amplifier.
7. Send preliminary quotation to the customer for the repair price.
8. Engineering decisions for step-by-step tests action for
   a. Additional tests needed for failure root cause verification
   b. upgrades if needed
   c. Component replacement needed for fixing the amplifier.
   d. Verification test
   e. ESS, Need to send the modules out for vibration per the specification.
   f. ATR test
   g. Failure reports
   9. Visual inspection before cover assembly.
10. Send final quotation to customer.
11. Receiving PO for the RMA module repair.
12. Approved for shipment.
13. Packaging and shipping instructions
    1. Put the do not open sticker
    2. Put protective cup on the RF and DC connectors.
    3. Put the amplifier in the antistatic bag.
4. Pack the unit in the carton package.

Important Note:
At any stage or time all actions need to be documented for the finding by pictures and writing. All decision and requirements must be followed.

Twik Engineering - Internal Use Only
This document contains proprietary information that shall be distributed, routed or made available only within Twik Engineering except with written permission of Twik Engineering.

Appendix 1
RMA Product Traveler-Amplifier

Part No:	TX-450	Customer:	Northwind Avionics		MRR REPORT #
RMA Number:	261001	Product S/N:	24110387		260543
Internal S/N:		Date:	11/06/2026		

Step No	Work	Description	Reference	N/Y	Responsible Department	Initial/ Signature	Date
1.	Mark S/N on chassis body	S/N Maintained	MOP # 19	Y	QC		
2.	**QC Inspect for damage and photograph**			Y	QC		
3.	Record Customer Complaint		MRR	Y	QC		
4.	Verify Customer Complaint and Record the finding	Diagnostic, test		Y	QC Test Engineer		
5.	Visual check after opening the cover, Record the finding			Y	QC		
6.	Diagnostic test to find which parts need to be replaced	Write the parts that need to be replaced at Table 1. One part for each line		Y	Test Engineer Engineering		
7.	Implement the required changes Updates	Revision and changes Updates	ECN	Y	Production Test Engineer		
8.	Replace Record parts as instructed by test engineer	Replace damaged parts	Table 1 & Assembly document	Y	Production Assembler		
9.	Electrical Testing	In case of replacing device need to check open or short.		Y	Production Assembler		
10.	Full visual Check after repair and update	Workmanship FOD	IPC-A- 610G CLASS 3 &	Y	Quality Control		
11.	Electrical Testing	In case of replacing device needs to check open or short.	-According to interaction	Y	Production Test Engineer		
12.	Biasing	Biasing each stage, record biasing current. Complete biasing tables.	-According to interaction	Y	Production Test Engineer	.	

Twik Engineering - Internal Use Only
This document contains proprietary information that shall be distributed, routed or made available only within Twik Engineering except with written permission of Twik Engineering.

13.	Tuning	Test to meet all RF and DC Function Test specifications.	- Specification - ATR	Y	Production Test Engineer		
14.	ESS	Overnight temperature cycles		Y	Test Engineer		
15.	SSG test & Biasing after ESS	Check model Biasing and re bias if necessary	According to interaction	Y	Test Engineer		
16.	Full ATR	Check: Electrical RF and DC test		Y	Production, Test Engineer		
17.	Final production Inspection and preparation before cover attachment	Check: Electrical wiring; RF connection and all internal components, solder joints, cleanses of the board. Add RTV for enforcement	- Assembly procedure - Mechanical drawings	Y	Production		
18.	Final Inspections before cover attachment	Check all soldering connections and wiring for flux residue. FOD	-IPC-A-610G CLASS 3	Y	Quality control		
19.	Cover closing	Attach absorber material to the cover. Close the cover with appropriate screws.	- Assembly procedure - Mechanical drawings	Y			
20.	Final Test	Test Go/No Go if any failure occurs during this stage, you must return to step # 6.		Y			
21.	Quality Control	QC 4 check		Y			
22.	Packing	Final Inspection / Verification/Tracking.	Packing procedure	Y	Production		
23.	Final inspection before shipment	Check the packing and documentation.		Y	Quality Control		

Final Approved by	**Date:**	**Sign**
Engineering Manager		
Production ETM Manager		
QC		

Revision History

Revision	Reason to Change	Date	Authorized by
00	Initial Issue		

Twik Engineering - Internal Use Only
This document contains proprietary information that shall be distributed, routed or made available only within Twik Engineering except with written permission of Twik Engineering.

Table 1 – Defective and tuning parts replacement						
Step #	**Issue/Description**	**P/N**	**Reference designator**	**Name**	**Date**	**Remarks**
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						
9.						
10.						
11.						
12.						
13.						
14.						
15.						
16.						
17.						
18.						

Twik Engineering - Internal Use Only
This document contains proprietary information that shall be distributed, routed or made available only within Twik Engineering except with written permission of Twik Engineering.